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CAPSTONE PROJECT 1 TITLE PROPOSAL No. 1

INTE1031 – Capstone Project 1

Group Name:_____5i_____ **Date of Presentation:**___March 4, 2026__

Position	Name	Signature
Project Manager	Loueala Jean H. Paquio	
System Analyst	John Bryan T. Gamboa	
Programmer	Felicite S. Astodillo	
Asst Programmer/System Designer	Jesie Marie D. Pagado	
Quality Assurance	John Greg A. Gonzales	

Proposed Thesis Title	***ABLEWORK: A WEB-BASED EMPLOYMENT ASSISTANCE SYSTEM FOR MATCHING PERSONS WITH DISABILITIES TO APPROPRIATE WORK OPPORTUNITIES***
Research Agenda	Information Systems and Assistive Technology
Research Gap	Many existing studies employment systems and online job portals, but they mostly focus on general job matching for everyone. Only a few studies focus on web-based platforms made especially for persons with disabilities (PWDs). Most existing systems do not fully consider different types of disabilities, individual skills, physical or mental limitations, and needed workplace support when matching jobs. Accessibility features are also often missing. Because of this, there is still a gap in creating an inclusive web-based employment system that properly helps PWDs find jobs that match their abilities and needs.
Overview of the Current State of the Technology	In Bacolod City, many persons with disabilities rely on general online job portals or traditional employment services to look for work. These platforms usually provide job listings and application forms but do not take into account the specific abilities, needs, or limitations of users with disabilities. Because of this, finding a suitable job can be slow, frustrating, and sometimes ineffective. Existing tools rarely provide personalized guidance or match jobs based on disability type, skills, or required workplace accommodations. This can lead to missed opportunities or jobs that are not fully appropriate. To improve accessibility and employment outcomes, there is a need for a web-based system in Bacolod City that connects persons with disabilities to suitable work opportunities, making the job search process faster, easier, and more inclusive.
Purpose Statement	The purpose of this study is to develop a web-based employment assistance system, ABLEWORK, that helps persons with disabilities in Bacolod City find jobs suited to their abilities and needs. The system aims to match users to appropriate work opportunities based on their skills, type of disability, and required workplace accommodations, providing an accessible, inclusive, and user-friendly platform. By doing so, the study seeks to improve employment outcomes, reduce barriers, and support the inclusion of persons with disabilities in the workforce.
Supporting Literature	Recent studies emphasize the importance of structured and inclusive employment systems for persons with disabilities. The study *"Strategies for Supporting Disability Inclusive Employment in the Future of Work"* highlights the need for adaptive workplace practices, technological support, and inclusive hiring strategies to ensure that persons with disabilities can actively participate in the evolving labor market. It stresses that digital tools and coordinated support systems play a critical role in reducing employment barriers and promoting long-term workforce participation (Hossain et al., 2025). Similarly, the work on supported employment and job design explains how carefully aligning job tasks with an

	individual's strengths, abilities, and accommodation needs increases job retention and performance among persons with disabilities. It emphasizes structured job matching and workplace adjustments as key components of successful inclusive employment (Wehman, 2025). These studies support the development of ABLEWORK, as the proposed system applies similar principles by matching persons with disabilities to suitable work opportunities based on their skills, type of disability, and workplace requirements. By integrating inclusive employment strategies and structured job alignment into a digital platform, the study contributes to promoting accessible and sustainable employment opportunities for persons with disabilities.
Supporting Systems/Studies	Recent system-focused studies explore both the challenges persons with disabilities face in the employment process and the potential of technology to improve recruitment outcomes. An action research study on employability assessment found that persons with disabilities encounter significant barriers in job applications, including limited information about suitable roles, lack of tailored support, and difficulty matching their skills to employer expectations; participants also described various coping mechanisms they use to navigate these challenges (Balala et al., 2024). Another study on inclusive hiring through technology demonstrated how a specialized recruitment platform can support individuals with intellectual disabilities by using digital tools to streamline job matching and provide accessible recruitment interfaces (Marini, Graf, & Millington, 2024). These findings reinforce the rationale for ABLEWORK, as the proposed employment assistance system addresses documented employability challenges by leveraging technology to match persons with disabilities to appropriate work opportunities. By combining structured job compatibility assessment with an accessible, technology-enabled platform, the study aims to reduce barriers and enhance inclusive hiring outcomes for users.
References	Balala, K. A. M., Claveria, H. S., Favia, J. M., Quintana, M. F. T., Santural, R. L., Quijano, E. D., Estrella, P. J. A., & Yap, R. S. (2024). An action research on employability assessment, challenges, and coping mechanism of persons with disabilities (PWDs) in applying for a job. International Journal of Research and Innovation in Social Science. https://rsisinternational.org/journals/ijriss/articles/an-action-research-on-employability-assessment-challenges-and-coping-mechanism-of-persons-with-disabilities-pwds-in-applying-for-a-job/ Hossain, S., Nasir, K., Carolli, A., Gignac, A. A. M., & Van Eerd, D. (2025). *Strategies for supporting disability-inclusive employment in the future of work*. Journal of Occupational Rehabilitation. https://link.springer.com/article/10.1007/s10926-025-10339-7 Marini, I., Graf, N. M., & Millington, M. J. (2024). *Inclusive hiring through technology: A recruitment platform for individuals with intellectual disabilities*. Procedia CIRP, 128, 31–36. https://www.sciencedirect.com/science/article/pii/S2212827124006425 Wehman, P. (2025). Supported employment and job design (in Inclusive Job Design, pp. 7.3). Springer. https://link.springer.com/chapter/10.1007/978-3-031-96315-5_7
Objectives	The main objective of this study is to design and develop ABLEWORK, a web-based employment assistance system that helps persons with disabilities in Bacolod City find jobs that match their skills, abilities, and type of disability. Specifically, this study aims to: • To create a user profiling system that records the type of disability, skills, qualifications, and work preferences of persons with disabilities. • To develop a job-matching mechanism that aligns users' abilities with workplace requirements. • To provide an accessible and user-friendly platform for job searching and application. • To connect persons with disabilities with employers offering inclusive job opportunities.

	• To reduce employment barriers by recommending suitable and meaningful work opportunities. • To evaluate the system's usability, accessibility, and effectiveness in assisting users.
Benefits/ Significance of the Study	This study is significant for the following stakeholders: **Persons with Disabilities (PWDs).** The study can benefit PWDs by providing a web-based platform that helps them find jobs suited to their skills, abilities, and specific needs. By matching users to appropriate work opportunities based on their disability type, individual skills, and workplace accommodations, the system can make the job search process faster, easier, and more effective. This can increase employment opportunities, build confidence, and support greater independence among PWDs. **Employers.** Employers can benefit from the system by accessing a pool of qualified candidates whose abilities and skills match the job requirements. The platform can help reduce mismatches between job roles and employee capabilities, making hiring more efficient and promoting inclusive workplaces. **Society & Community.** The study contributes to creating a more inclusive and equitable society. By helping PWDs gain meaningful employment, it promotes social inclusion, reduces unemployment, and encourages community awareness about the abilities and contributions of persons with disabilities. It also supports sustainable workforce development by ensuring that all members of the community have equal access to economic opportunities. **Researchers.** This study can serve as a reference for researchers interested in assistive technology, inclusive information systems, and employment support platforms. It provides insights into system design, job-matching mechanisms, and accessibility considerations for persons with disabilities. **Future Researchers.** Future researchers may use this study as a foundation for improving or expanding employment assistance systems. They may explore advanced matching algorithms, mobile-based solutions, integration with government employment programs, or the inclusion of additional accessibility features to further enhance system effectiveness and inclusivity.
Scope	This study focuses on developing a web-based employment assistance system, a ABLEWORK, for persons with disabilities (PWDs) in Bacolod City. The system will help users find jobs that match their skills, abilities, type of disability, and workplace needs. It will provide job listings, matching features, and guidance to make the job search easier and more accessible. The study is limited to jobs and employers registered in the system within Bacolod City. The main focus is on creating an inclusive and user-friendly platform that connects PWDs to suitable work opportunities.
Limitations	This study is limited to persons with disabilities (PWDs) in Bacolod City who use the ABLEWORK system. The system only includes job listings and employers registered within Bacolod City, so it cannot provide opportunities outside the city. It focuses only on web-based access, which means users need a device and internet connection to use the platform. The study does not cover other types of employment support, such as in-person training, career counseling, or workshops that help PWDs find jobs. Finally, the system provides guidance and job matching but does not guarantee that users will get employed.
Target Users	The target users of this study are persons with disabilities (PWDs) living in Bacolod City who are seeking employment opportunities. The system is designed to assist PWDs of different disability types and levels of ability by providing job listings, matching features, and guidance tailored to their skills and workplace needs. The platform is also intended for employers in Bacolod City who want to hire qualified candidates with disabilities, helping them find workers whose abilities match the job requirements.

UN's Sustainable Development Goals (SDGs)	SDG 8 – Decent Work and Economic Growth SDG 10 – Reduced Inequalities
How your research will contribute to the field.	This research will help the field by showing how a web-based system can help persons with disabilities (PWDs) find suitable jobs. It provides a model for matching jobs based on skills, abilities, and disability needs, which is not common in most job websites. The study can guide future developers and researchers to create more inclusive employment systems that reduce barriers and give PWDs better access to work opportunities.
STATUS	☐ Approved ☐ Rejected
Comments/ Suggestions/ Recommendations	
Capstone Panelist	Name & Signature

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CAPSTONE PROJECT 1 TITLE PROPOSAL No. 2

INTE 1031 – Capstone Project 1

Group Name:_____5i_____ **Date of Presentation:**___March 4, 2026___

Position	Name	Signature
Project Manager	Loueala Jean H. Paquio	
System Analyst	John Bryan T. Gamboa	
Programmer	Felicite S. Astodillo	
Asst Programmer/System Designer	Jesie Marie D. Pagado	
Quality Assurance	John Greg A. Gonzales	

Proposed Thesis Title	**FOODCONNECT: A DONATION MANAGEMENT SYSTEM FOR ZERO HUNGER EFFORTS**
Research Agenda	Mobile and Web-Based Donation Systems
Research Gap	Many studies have explored the role of digital platforms in food donation and waste reduction. For example, research on food-sharing applications like Olio and Too Good To Go demonstrates how mobile platforms can connect surplus food providers with consumers to reduce food waste. Additionally, studies on food bank management systems highlight the importance of database-driven solutions for tracking donations and inventory. However, most existing platforms primarily focus on either commercial surplus food redistribution or localized food bank operations. There is limited research on an integrated donation management system specifically designed to connect individual donors, organizations, and beneficiaries in a localized Philippine context while incorporating real-time tracking, verification, and analytics features. Therefore, a research gap exists in the development of a comprehensive, scalable, and community-centered donation management system that leverages mobile and web technologies to improve efficiency, transparency, and impact measurement in food donation programs.
Overview of the Current State of Technology	In many areas in the Philippines, food donation efforts are still managed through manual and informal methods such as phone calls, social media messages, and paper-based records. While these approaches allow food to reach communities in need, they are often slow, difficult to monitor, and prone to errors. Some mobile applications help reduce food waste, but most are limited in scope and do not provide a centralized system for tracking donations and distribution. As a result, coordination gaps and inefficiencies still exist. There is need for an integrated digital platform that enables real-time tracking, organized data management, and efficient matching donors and beneficiaries to support Zero Hunger efforts.
Purpose Statement	The purpose of this thesis study is to design and develop FOODCONNECT, a donation management system that streamlines food donation processes by connecting donors, organizations, and beneficiaries through a centralized digital platform. The system aims to improve coordination, enhance transparency, enable real-time tracking, and support efficient food distribution to contribute to Zero Hunger efforts in the Philippines.

Supporting Literature	Digital platforms have emerged as innovative solutions in reducing food waste and promoting sustainable food systems. In *Understanding Food Waste-Reducing Platforms: A Mini-Review* (Oroski & Silva, 2022), food waste-reducing platforms (FWRPs) are described as technological systems that connect surplus food supply with demand through donation or commercialization. The study explains that these platforms help monitor food waste, generate valuable data, and enable surplus sharing among stakeholders. However, the authors emphasize that technology alone is not sufficient; successful implementation requires alignment with sustainability principles, including environmental, social, and economic benefits, as well as trust-building and behavioral change among users. In addition, Papargyropoulou et al. (2014) proposed a comprehensive framework for addressing food waste by interpreting the waste hierarchy within the context of food systems. The study examined the causes of food waste across the global food supply chain and emphasized the importance of adopting a sustainable production and consumption approach. The proposed food waste hierarchy prioritizes prevention by minimizing food surplus and avoidable food waste as the most desirable solution. The second priority involves redistributing surplus food to groups affected by food poverty, followed by converting food waste into animal feed. The framework integrates environmental, social, and economic dimensions of sustainability, offering a holistic approach to food waste management. These studies provide strong theoretical foundations for the present research. The proposed FOODCONNECT system aligns with the food waste hierarchy by prioritizing the redistribution of surplus food before disposal. By integrating digital coordination, real-time notifications, and data monitoring, the system supports prevention and redistribution strategies while contributing to environmental sustainability and hunger reduction efforts.
Supporting Systems/Studies	**FeedBack: Food Donation Application in Selangor** Shankarvelu et al. (2023) developed FeedBack, a food donation application designed to improve the distribution and verification process of food aid in Selangor, Malaysia. The system was created in response to unequal food distribution during crises such as floods and the COVID-19 pandemic. It provides a centralized platform that connects donors, NGOs, volunteers, and recipients to ensure proper coordination. The application emphasizes cybersecurity features to protect user data and incorporates survey-based evaluation to assess public acceptance. By improving communication between stakeholders, the system reduces food wastage and ensures that food donations reach areas with greater need. This system supports the present study because it demonstrates how a centralized digital platform can address inefficiencies in food distribution. Similar to FeedBack, the proposed FOODCONNECT system aims to improve coordination, transparency, and donation tracking. However, FOODCONNECT is specifically designed to address food insecurity challenges within the Philippine context, incorporating localized features suited for community-based food donation management. **SEVA: A Real-Time Solution for Food Donation and Hunger Alleviation** Kumar and Raj (2021) introduced SEVA, a real-time food donation system developed in India to reduce hunger and food waste. The application connects surplus food sources such as households, restaurants, NGOs, and event organizers with individuals experiencing food insecurity. SEVA utilizes modern technologies, including real-time communication, database systems, secure login and registration features, and cloud-based solutions to ensure efficient food redistribution. The system also promotes sustainability by minimizing food waste and reducing environmental impact. This study is relevant to the present research as it demonstrates the effectiveness of integrating modern information systems to address hunger and food waste. Like SEVA, FOODCONNECT aims to serve as a digital bridge between donors and beneficiaries. However, while SEVA focuses on hunger alleviation in India, the proposed system adapts the concept to the Philippine setting and emphasizes structured donation management, transparency, and monitoring mechanisms to enhance food distribution efficiency.
References:	Kumar, V., & Raj, S. (2021). SEVA: A real-time solution for food donation and hunger alleviation. Lovely Professional University, Phagwara, India.

	Papargyropoulou, E., Lozano, R., Steinberger, J. K., Wright, N., & Ujang, Z. (2014). *The food waste hierarchy as a framework for the management of food surplus and food waste*. Journal of Cleaner Production, 76, 106–115. https://www.sciencedirect.com/science/article/abs/pii/S0959652614003680?via%3Dihub Oroski, F. de A., & Silva, J. M. da. (2022). *Understanding food waste-reducing platforms: A mini-review*. Waste Management & Research, 41(4). https://doi.org/10.1177/0734242X221135248 Shankarvelu, V. A. L., Hakimi, H., Marlia, Z., & Octaviani, D. (2023). *Food donation application to improve the distribution and verification process within Selangor: FeedBack*. Journal of Applied Technology and Innovation, 7(3).
Objectives	The main objective of this study is to develop the FOODCONNECT system using suitable technologies for web and mobile access. Specifically, this study aims to: • To implement secure user authentication for donors, organizations, and volunteers. • To integrate location-based features for efficient matching and coordination. • To design and manage a database system for storing and tracking donation records. • To ensure real-time updates and notifications for donation status. • To create a user-friendly interface accessible on both web and mobile devices.
Benefits/ Significance of the Study	This study is significant for the following stakeholders: **Donors (individuals, households, restaurants, food businesses):** FOODCONNECT provides an easy-to-use digital platform for posting surplus food with real-time details (quantity, type, location, photos) and instant notifications to nearby organizations. Donors gain transparency through tracking features (e.g., donation status updates and impact dashboards showing kg redistributed or beneficiaries reached), building confidence and encouraging regular contributions over informal methods like social media or calls. This reduces personal food waste while enabling meaningful support for hunger relief in Bacolod City communities. **Organizations (NGOs, Local Government Units - LGUs, and Department of Social Welfare and Development - DSWD):** NGOs, LGUs, and DSWD benefit from a centralized system that streamlines real-time access to donations, automated matching with needy groups, verification for food safety, and efficient coordination for distribution. NGOs involved in grassroots hunger programs (e.g., daily meals for children in squatter areas or crisis aid) can minimize delays and wastage. LGUs, often partnering in local relief and community feeding, gain better resource allocation and data for targeting aid. For DSWD—leading the Walang Gutom Program with food redemptions, stamps, and packs in Bacolod City—the platform enhances partnerships, supports timely access to nutritious food for low-income households, reduces administrative burdens from manual processes, and provides insights to strengthen national and regional hunger alleviation efforts. **Volunteers:** Volunteers facilitating matching, pickup, or distribution gain tools for real-time alerts, location-based coordination, and status tracking, making their efforts more organized and effective. The platform reduces coordination challenges in manual setups, allows quick response to donations, and offers visibility into impact (e.g., successful deliveries to feeding programs), motivating continued participation and enabling scalable support in local hunger initiatives across Bacolod City. **Society & Community (families, shelters, feeding programs, and similar vulnerable groups):** FOODCONNECT bridges surplus food to those in need through transparent, efficient matching and notifications, ensuring quicker and more equitable access to donations for low-income families, shelters (e.g., orphanages or crisis homes), community feeding programs, and other vulnerable groups facing involuntary hunger. In localized settings like Bacolod City, it helps reduce food insecurity by minimizing waste, supporting nutrition for children and families in underserved areas, and fostering resilience during daily hardships or crises—contributing to stronger, more sustainable communities aligned with Zero Hunger goals.

	Researchers: This study provides a foundation for researchers to analyze the effectiveness of digital platforms for social impact, food redistribution, and community-based hunger solutions. Researchers can study system efficiency, user experience, and measurable social benefits to contribute knowledge to the field of technology-driven social welfare initiatives. **Future Researchers:** Future researchers may build on this study by exploring long-term effects of the platform, expanding its features, or adapting the model for other cities and regions. They can investigate additional strategies for reducing food waste, improving hunger alleviation, and strengthening community resilience, further advancing sustainable development and innovative social support systems.
Scope	This study focuses on the design, development, and evaluation of FOODCONNECT, a web and mobile-accessible donation management system for food redistribution in the Philippines, with initial localization to Bacolod City. It includes user registration and profile management for donors (individuals, households, restaurants, supermarkets), organizations (NGOs, LGUs, DSWD representatives aligned with programs like Walang Gutom), and volunteers; features for posting surplus food with details (type, quantity, expiration, location, photos), real-time automated matching and notifications, donation status tracking, verification (admin approval, ratings), location-based services for nearby connections, basic impact analytics dashboards (e.g., kg redistributed, beneficiaries served), secure authentication, and push/email alerts. Development uses standard responsive web/mobile technologies with database support. Testing covers functionality, usability, performance, and user satisfaction through unit/integration testing, user acceptance testing, and feedback from sample local users representing the target groups.
Limitations	The study is limited to software development and evaluation within the capstone timeline and resources. It does not include hardware integration, physical logistics, or transport coordination (users must arrange pickups/deliveries independently), full national deployment, or long-term operational maintenance, or advanced features such as AI-based food quality assessment, blockchain traceability, or offline functionality. The platform relies on user-input data and internet/smartphone access, so inaccuracies in donations or limited reach in low-connectivity areas may occur. Initial testing is localized to Bacolod City with a limited sample size of users, not long-term real-world impact or scalability across the Philippines. Food safety ultimately depends on user compliance and verification processes, not guaranteed physical inspections.
Target Users	The primary target users of FOODCONNECT are individual donors (households, community members willing to share surplus home-cooked or purchased food); food businesses and organizations (restaurants, supermarkets with excess/perishable items); NGOs, charities, and local government units involved in food distribution and hunger relief; beneficiaries (individuals/families experiencing food insecurity, or community feeding programs); and volunteers or coordinators who facilitate matching and logistics.
UN's Sustainable Development Goals (SDGs)	SDG 2 - Zero Hunger SDG 12 – Responsible Consumption and Production
How your research will contribute to the field.	This research fills a key gap in mobile and web-based donation systems by developing FOODCONNECT—a localized, community-centered platform for the Philippines that integrates individual donors, NGOs, LGUs, DSWD, and volunteers with real-time tracking, verification, and impact analytics in a non-commercial donation model. Unlike most existing studies focused on commercial surplus apps or non-Philippine contexts, it provides empirical insights from Bacolod City testing, demonstrates improved coordination, transparency, and efficiency in food redistribution to vulnerable groups, and offers a scalable prototype aligned with SDG 2 (Zero Hunger). The work advances academic understanding of technology-driven hunger alleviation in developing settings and lays the foundation for future regional expansions and policy integrations (e.g., with DSWD programs).
STATUS	☐ Approved ☐ Rejected
Comments/ Suggestions/ Recommendations	
Thesis Panelist	Name & Signature

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THESIS TITLE PROPOSAL No. 3

COSC1048 – Methods of Research (Thesis 0)

Group Name: 5i **Date of Presentation:** March 4, 2026

Position	Name	Signature
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Quality Assurance	John Greg A. Gonzales	

Proposed Thesis Title	**GeoCropGuard: A GEO-TAGGED CROP DISEASE MONITORING AND OUTBREAK MAPPING SYSTEM**
Research Agenda	Computer Vision and Geographic Information Systems
Research Gap	Many studies have explored crop disease detection using image processing, remote sensing, and vegetation indices, focusing mainly on improving classification accuracy under controlled datasets. However, there has been limited attention on integrating geo-tagged disease detection with real-time outbreak mapping for field-level monitoring. Existing systems typically detect diseases from images but do not provide location-based analytics, spatial clustering of infections, or automated outbreak visualization that can support early intervention by farmers and agricultural agencies. In addition, most studies rely on static datasets and lack mobile, user-friendly platforms that allow farmers to capture images in the field and instantly contribute to a centralized disease surveillance map. Therefore, there is a need to develop a geo-tagged crop disease monitoring and outbreak mapping system that combines image-based detection, GPS data, and real-time spatial analysis to enable faster decision-making and targeted disease control.
Overview of the Current State of Technology	Current crop disease monitoring technologies use AI-based image recognition, remote sensing, and vegetation indices (e.g., NDVI) to detect plant diseases and assess crop health. Deep learning models can classify diseases from leaf images, while satellite and drone imagery help identify stressed or infected areas for precision agriculture. These technologies improve early detection and reduce manual field inspection. However, most existing systems are designed for disease identification only and operate on static datasets or large-scale imagery. They do not commonly provide geo-tagged, field-level data collection, real-time outbreak mapping, or a centralized mobile platform for farmers to report and visualize disease spread. This limits timely decision-making and localized intervention, which the proposed GeoCropGuard system aims to address.
Purpose Statement	The purpose of this thesis study is to develop GeoCropGuard: A geo-tagged crop disease monitoring and outbreak mapping system that allows farmers to report diseases in real-time, save data in a centralized database, and visualize the affected area on a map. It is intended to provide faster response, more informed decisions and better crop disease management.

Supporting Literature	Geographic Information Systems (GIS) and spatial technologies have become essential tools in modern agriculture for monitoring crop health and supporting evidence-based decision-making. In *Application of GIS in Agriculture in Promoting Evidence-Informed Decision Making for Improving Agriculture Sustainability: A Systematic Review* (Mathenge, Sonneveld, & Broerse, 2022), the authors reviewed how GIS and Remote Sensing (RS) applications enhance agricultural sustainability through crop yield estimation, soil fertility assessment, cropping pattern monitoring, drought assessment, and pest and crop disease management. The study emphasizes that integrating spatial data into agricultural practices improves decision-making by providing timely, location-specific information, enabling better resource allocation and operational efficiency. Similarly, Wang, Chen, and Dong (2013) investigated early-warning technologies for crop diseases and insect pests, focusing on the limitations of traditional monitoring methods such as field visits, sampling, and analysis. Their study highlights how IoT-enhanced early-warning systems improve the accuracy and timeliness of disease detection. By reviewing mainstream warning algorithms and proposing technology integration, the study underscores the importance of digital solutions in providing farmers with actionable, real-time information to respond to pest and disease threats. These studies provide strong theoretical foundations for the present research. The proposed GeoCropGuard system builds on these principles by combining GIS-based geotagging and mapping with real-time monitoring of crop diseases. By providing timely alerts and spatially referenced data, the system enables farmers and authorities to make informed decisions, mitigate crop losses, and enhance overall agricultural sustainability.
Supporting Systems/Studies	Recent technological advancements demonstrate the growing role of artificial intelligence and geospatial systems in agricultural disease monitoring. Lee and Yun (2023) developed a deep learning model that utilizes sequential environmental factors such as temperature, humidity, dew point, and CO_2 levels to predict the probability of crop disease occurrence. The model achieved high accuracy in forecasting disease risk, emphasizing the strong relationship between environmental conditions and disease development. Their study highlights the effectiveness of data-driven approaches in supporting early warning systems and improving crop management strategies. Similarly, the Distance Diagnostic and Identification System (DDIS) developed by the University of Florida enables users to submit plant disease and pest samples through geo-tagged images. By extracting GPS location and timestamp data from EXIF metadata, the system allows mapped tracking of pest and disease occurrences for monitoring and early detection. It facilitates collaboration between field agents and experts while maintaining an archived database for research and surveillance purposes. Although both systems contribute significantly to agricultural disease monitoring, certain limitations remain. The deep learning model by Lee and Yun (2023) focuses primarily on environmental risk prediction and does not incorporate geo-tagged field reports, spatial outbreak visualization, or farmer-based mobile inputs. On the other hand, DDIS is largely designed for expert-based diagnosis and requires authorized personnel, lacking automated computer vision detection, real-time alerts, and interactive outbreak mapping accessible to farmers. The proposed GeoCropGuard system extends these existing approaches by integrating environmental risk prediction, AI-based image detection, geo-tagged farmer reporting, and real-time GIS visualization into a unified mobile-accessible platform. By combining predictive analytics with spatial outbreak monitoring, the system aims to provide more comprehensive, practical, and accessible decision support for farmers and agricultural agencies.
References:	Elbeltagi, A. (2025). *Deep learning and computer vision in plant disease detection: A comprehensive review of techniques, models, and trends in precision agriculture.* Artificial Intelligence Review, 58(3), 92. https://link.springer.com/article/10.1007/s10462-024-11100-x Lee, S., & Yun, C. M. (2023). *A deep learning model for predicting risks of crop pests and diseases from sequential environmental data*. Plant Methods, 19, 145. https://link.springer.com/article/10.1186/s13007-023-01122-x

	Mathenge, M., Sonneveld, B. G. J. S., & Broerse, J. E. W. (2022). *Application of GIS in agriculture in promoting evidence-informed decision making for improving agriculture sustainability: A systematic review.* Sustainability, 14(16), 9974. https://www.mdpi.com/2071-1050/14/16/9974 Wang, D., Chen, T., & Dong, J. (2013). *Research of the early warning analysis of crop diseases and insect pests.* In S. T. Yang, A. L. Goel, & J. B. Luo (Eds.), *Computer and Computing Technologies in Agriculture VII* (pp. 177–187). Springer. https://link.springer.com/chapter/10.1007/978-3-642-54341-8_19 Yan, J., Wu, H., Diao, Z., Miao, Y., Zhang, B., & Zhao, C. (2025). Recent developments and applications of crop disease detection, prediction, and early warning: A review. Engineering. https://doi.org/10.1016/j.eng.2025.10.032 Xin, J. (2009). Tracking pests and plant diseases using geo-tagged digital images. University of Florida, IFAS Extension. https://ddis.ifas.ufl.edu/gps/gpsphoto.pdf
Objectives	The main objective of this study is to develop GeoCropGuard, a geo-tagged crop disease monitoring and outbreak mapping system that enables real-time detection, reporting, and spatial visualization of crop diseases. Specifically, this study aims to: • To implement mobile-based image capture and automated analysis for early crop disease detection. • To integrate geo-tagging and mapping for real-time visualization of disease outbreaks. • To provide real-time reporting tools for farmers, LGUs, and the Department of Agriculture. • To design a centralized database for storing disease records and outbreak data. • To create a user-friendly interface accessible on mobile and web platforms.
Benefits/ Significance of the Study	**Farmers:** GeoCropGuard helps farmers detect crop diseases early using mobile image capture with automated analysis and location tagging. This enables faster treatment, reduces crop losses, lowers production costs, and improves yield and income. **Local Government Units (LGUs):** The system provides real-time outbreak maps and localized reports that help LGUs monitor disease spread, plan targeted interventions, allocate resources efficiently, and implement timely agricultural support programs. **Department of Agriculture (DA):** GeoCropGuard offers a centralized geo-tagged database for large-scale surveillance, policy planning, and evidence-based decision-making. It supports early warning systems, improves coordination with LGUs, and strengthens national crop protection and food security initiatives. **Researchers:** This study provides a foundation for researchers to analyze the use of technology in crop disease detection, precision agriculture, and data-driven agricultural management. Researchers can study system performance, user adoption, and the impact on crop health and yield. **Future Researchers:** Future researchers may expand this study by integrating additional technologies, adapting the system for other crops or regions, or assessing long-term effects on agricultural productivity and food security. Their work can contribute to improving sustainable farming practices and national crop protection strategies.
Scope	This study focuses on the design and development of GeocropGuard: A Geo-Tagged Crop Disease Monitoring and Outbreak Mapping System intended for agricultural use in the Philippines, with its initial implementation and pilot testing conducted in Bacolod City, Negros Occidental. The system will utilize GPS-enabled photo documentation and geotagging technology to monitor, record, and visualize crop disease occurrences. Farmers and agricultural personnel will be able to capture or upload images of infected crops using mobile devices. The system

	will automatically extract geographic metadata such as latitude, longitude, date, and time from the uploaded images and store the information in a centralized database. Reported disease cases will be displayed on an interactive digital map. During the pilot phase, mapping will focus on agricultural areas within Bacolod City to identify affected barangays, visualize disease distribution, and detect potential outbreak clusters. The system architecture, however, will be designed to support future expansion across other cities and provinces in the Philippines.
Limitations	This study is limited to the development and implementation of GeocropGuard in Bacolod City, Negros Occidental. The system depends on user-submitted geo-tagged photos, and inaccurate GPS signals, poor image quality, or incomplete data may affect mapping accuracy. The system does not include advanced AI-based disease detection, environmental sensor integration, or real-time predictive modeling. Data collected during the pilot phase may also be limited, which can affect outbreak analysis. Additionally, the effectiveness of the system depends on internet connectivity, smartphone availability, and the technical skills of users. Limited access to technology in some areas may affect timely reporting. These limitations define the boundaries of the study during its initial implementation in Bacolod City.
Target Users	**Farmers (Primary Users)** Smallholder and commercial farmers will be the main users of the system. They can use the mobile application to capture crop images, report diseases, and view outbreak maps in their area. This helps them take early action to prevent crop loss. **Local Government Units (LGUs)** – Agriculture Office Municipal and provincial agriculture offices can use the outbreak map for decision-making, resource allocation, and early warning alerts to affected farming communities. **Department of Agriculture (DA)** The DA can use aggregated data for regional disease surveillance, planning interventions, and developing pest and disease control programs.
UN's Sustainable Development Goals (SDGs)	SDG 2 – Zero Hunger SDG 9 – Industry, Innovation, and Infrastructure SDG 12 – Responsible Consumption and Production
How your research will contribute to the field.	This research will contribute to the field by addressing the gap between crop disease image detection and real-world agricultural monitoring systems. GeoCropGuard will integrate computer vision, geo-tagging, and real-time outbreak mapping into a unified mobile and web-based platform to enable field-level disease reporting and spatial analysis. Unlike existing studies that focus mainly on classification accuracy using static datasets, this study will propose a localized, farmer-centered surveillance system to be piloted in Bacolod City. It will enhance early detection, support data-driven decision-making for farmers and agricultural agencies, and offer a scalable prototype aligned with SDG 2 (Zero Hunger). The research will advance the practical application of computer vision and GIS technologies in smart agriculture within a developing-country context.
STATUS	☐ Approved ☐ Rejected
Suggestions/ Recommendations	
Thesis Panelist	Name & Signature